Cohort canonical evaluation — 30 agents

idea-01-lateral-attribution · 534 held-out segments · V0 yaw 0.01627 rad/s, CTE 254.26 m · generated 2026-06-01 00:23

Executive summary

AGENTS (OK / TOTAL)

30 / 30

EVAL POOL

534 segs

V0 YAW RMSE

0.01627

V0 CTE RMSE

254.26 m

WALL TIME

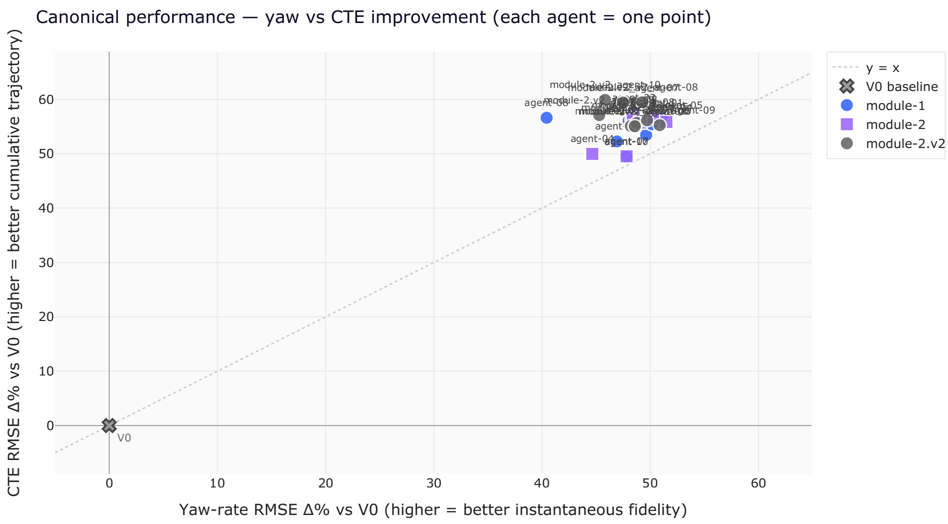
19.43s

Best yaw: `m2-agent-05` **+51.5%** · Best CTE: `module-2.v2_agent-10` **+59.9%**

Winning both KPIs ≥ +30% (30 agents): `m2-agent-05`, `m1-agent-01`, `module-2.v2_agent-09`, `m2-agent-08`, `m1-agent-09`, `module-2.v2_agent-05`, `m1-agent-07`, `module-2.v2_agent-08`, `m2-agent-10`, `module-2.v2_agent-04`, `m1-agent-02`, `m1-agent-05`, `m2-agent-01`, `module-2.v2_agent-06`, `module-2.v2_agent-02`, `m2-agent-03`, `m1-agent-03`, `m2-agent-02`, `m2-agent-09`, `m2-agent-06`, `module-2.v2_agent-01`, `m1-agent-04`, `m2-agent-07`, `m1-agent-10`, `module-2.v2_agent-07`, `m1-agent-06`, `module-2.v2_agent-10`, `module-2.v2_agent-03`, `m2-agent-04`, `m1-agent-08`

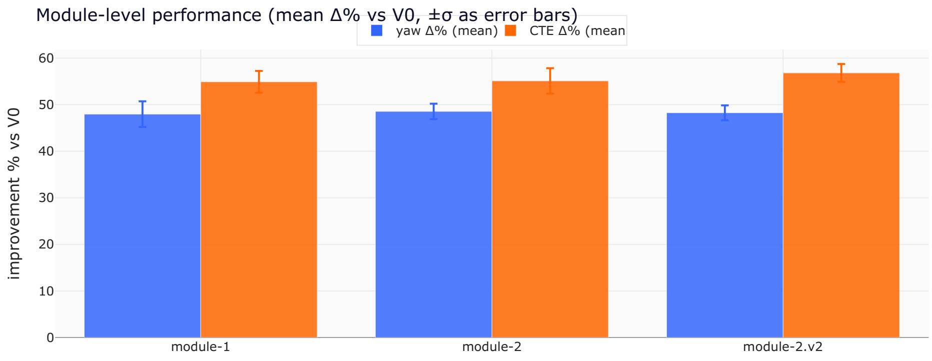
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is y = x: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

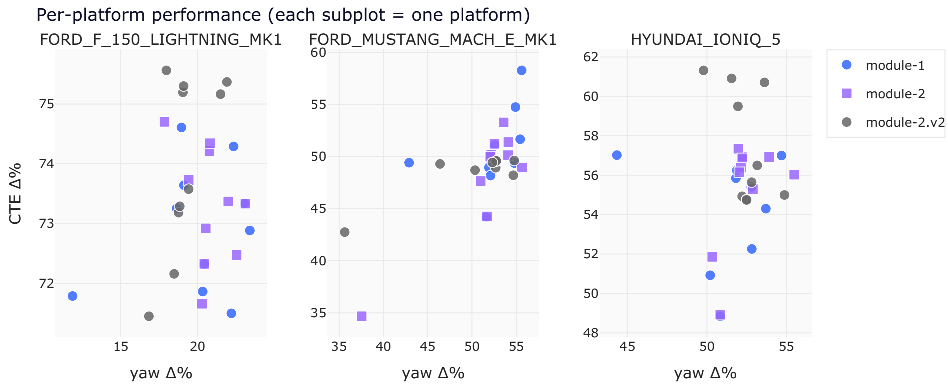
Performance grouped by which module shipped the agent. Mean Δ% over only the agents that reconstructed successfully; failures shown separately.



family	n ok / total	yaw Δ% (mean ± σ)	CTE Δ%
module-1	10/10	+48.0% ± 2.8%	+54.9
module-2	10/10	+48.6% ± 1.7%	+55.1
module-2.v2	10/10	+48.3% ± 1.6%	+56.8

3. Per-platform breakdown

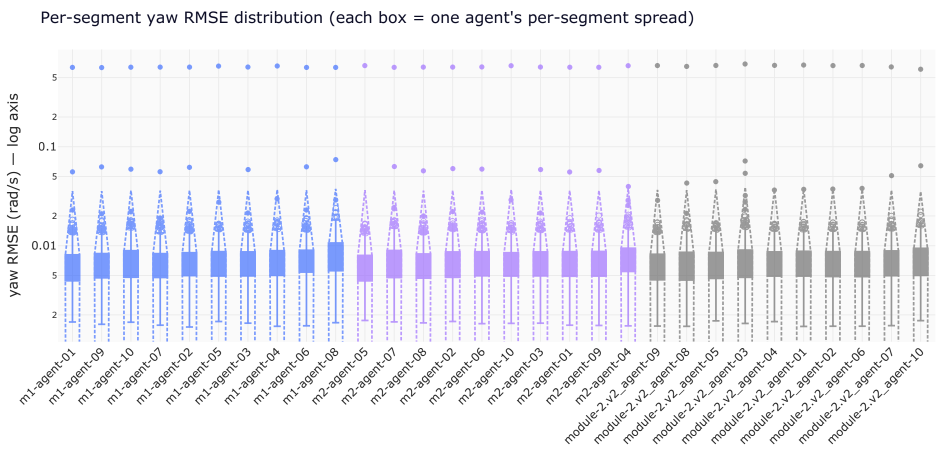
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw Δ% (mean ± σ)
FORD_F_150_LIGHTNING_MK1	30	+20.0% ± 2.3%
FORD_MUSTANG_MACH_E_MK1	30	+51.5% ± 4.7%
HYUNDAI_IONIQ_5	30	+52.1% ± 1.9%

4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status	yaw V0
m1-agent-01	module-1	ok	0.016274
m1-agent-02	module-1	ok	0.016274
m1-agent-03	module-1	ok	0.016274
m1-agent-04	module-1	ok	0.016274
m1-agent-05	module-1	ok	0.016274

agent	family	status	yaw V0
m1-agent-06	module-1	ok	0.016274
m1-agent-07	module-1	ok	0.016274
m1-agent-08	module-1	ok	0.016274
m1-agent-09	module-1	ok	0.016274
m1-agent-10	module-1	ok	0.016274
m2-agent-01	module-2	ok	0.016274
m2-agent-02	module-2	ok	0.016274
m2-agent-03	module-2	ok	0.016274
m2-agent-04	module-2	ok	0.016274
m2-agent-05	module-2	ok	0.016274
m2-agent-06	module-2	ok	0.016274
m2-agent-07	module-2	ok	0.016274
m2-agent-08	module-2	ok	0.016274
m2-agent-09	module-2	ok	0.016274
m2-agent-10	module-2	ok	0.016274
module-2.v2_agent-01	module-2.v2	ok	0.016274
module-2.v2_agent-02	module-2.v2	ok	0.016274

agent	family	status	yaw V0
module-2.v2_agent-03	module-2.v2	ok	0.016274
module-2.v2_agent-04	module-2.v2	ok	0.016274
module-2.v2_agent-05	module-2.v2	ok	0.016274
module-2.v2_agent-06	module-2.v2	ok	0.016274
module-2.v2_agent-07	module-2.v2	ok	0.016274
module-2.v2_agent-08	module-2.v2	ok	0.016274
module-2.v2_agent-09	module-2.v2	ok	0.016274
module-2.v2_agent-10	module-2.v2	ok	0.016274

6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
```

```
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
    "b": 0.9120045284217976,
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.00022067579452559 14,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "_default": {
    "a": 0.0,
    "b": 1.0,
    "c": 0.0
  }
}
```

m1-agent-06

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
```

```
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    },
    "m3": {
      "a": 0.9751496691433119,
      "b": 0.23839497273290036,
      "c": 0.00040828706448247888
    },
    "val_rmse": {
      "V0": 0.0267999381028469,
      "M1": 0.027439581036525294,
      "M2": 0.026420879339395156,
      "M3": 0.031903168689021696
    },
    "train_rmse": {
      "V0": 0.012685628695945105,
      "M1": 0.011389013310887795,
      "M2": 0.008745921115146504,
      "M3": 0.010386863485176999
    }
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "best": "M2_understeer",
    "m1": {
      "a": 0.8569324797501493
    },
    "m2": {
      "L_eff": 3.882198880475702,
      "K_us": 0.003439279945738158
    },
    "m3": {
      "a": 0.8772240685582882,
      "b": -0.031353830652592465,
      "c": -0.0036354381262862774
    },
    "val_rmse": {
      "V0": 0.01591104278075153,
      "M1": 0.011141896721881754,
      "M2": 0.0073696198273352906,
      "M3": 0.0104650790814036
    },
```

```
    "train_rmse": {
      "V0": 0.02018452928972004,
```

m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
```

```
  }
}
```

m1-agent-10

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred_
    "passthrough": true
  }
}
```

m2-agent-03


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
    "n_samples": 430990,
    "rmse_v0": 0.016327157090019005,
    "rmse_v1": 0.007867193883977044,
    "K_v1": 0.0043769531249999985,
    "rmse_v2": 0.00640773025816659,
    "K_v2": 0.003836687017912508,
    "a_v2": 0.9743406550510199,
    "b_v2": -0.001228592917784131
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.984,
    "n_samples": 615175,
    "rmse_v0": 0.013616820108542612,
    "rmse_v1": 0.013930807470187448,
```

```
    "K_v1": 0.000785400390624989,
    "rmse_v2": 0.0095149036095479,
    "K_v2": 0.0028742800060782272,
    "a_v2": 1.1998919734409932,
    "b_v2": 3.385609342398125e-05
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
    "L": 3.0,
    "n_samples": 2042270,
    "rmse_v0": 0.017084454875116246,
    "rmse_v1": 0.009100616867655459,
    "K_v1": 0.0042265624999999985,
    "rmse_v2": 0.008741269113389604,
    "K_v2": 0.0033859035164445724,
    "a_v2": 0.9707125684779052,
    "b_v2": 0.0005066770839987745
  }
}
```

m2-agent-07

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "tau": 0.05,
    "L_eff": 3.8588781722695416,
    "K_u": 0.00314039574481963,
    "delta_bias_rad": -0.001157331673169034,
    "rmse": 0.005592721659455879
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "tau": 0.08,
    "L_eff": 2.5486304264900417,
    "K_u": 0.0015312706121448161,
    "delta_bias_rad": 4.8217638587095315e-05,
    "rmse": 0.008347407485517062
```

```
  },
  "HYUNDAI_IONIQ_5": {
    "tau": 0.05,
    "L_eff": 3.163750474852756,
    "K_u": 0.0021607848879418314,
    "delta_bias_rad": 0.0005617097326308059,
    "rmse": 0.007713638558960502
  }
}
```

module-2.v2_agent-01

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a0": 0.9828241803279664,
    "a1": -0.012530358326866366,
    "a2": -7.476576241011569e-05,
```

```
    "b": -0.004437500352969203
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a0": 1.2437488124691014,
    "a1": -0.017093439455569934,
    "a2": -6.197874378671898e-05,
    "b": 0.00015242523943702585
  },
  "HYUNDAI_IONIQ_5": {
    "a0": 0.9650336037033849,
    "a1": -0.009921780292871708,
    "a2": -0.00019063509624679564,
    "b": 0.00200848042320952
  },
  "TESLA_MODEL_3": {
    "a0": 1.0,
    "a1": 0.0,
    "a2": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.00136,
    "delta_offset_rad": 0.0014,
    "tau_s": 0.0551
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.00027,
    "delta_offset_rad": -4e-05,
    "tau_s": 0.0804
  },
  "HYUNDAI_IONIQ_5": {
```

```
    "L": 3.0,
    "K": 0.00148,
    "delta_offset_rad": -0.00051,
    "tau_s": 0.0509
  }
}
```

module-2.v2_agent-05

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "yv0": 10.359025218583886,
    "v": -2.8512627155390877e-05,
    "d": -0.1421351518833939,
    "vd": -2.5140507093900077,
    "v2d": -0.004539666009077931 5,
    "d3": -0.9520178765297359,
    "vd3": -0.9563597630766577,
    "v2d3": -0.011640743270919 72,
    "sr": -0.043545329960739045,
    "vsr": -0.008335174282041887,
    "d_abs_d": 0.7154586232894494,
    "a_long": 0.0005002706723089025,
    "intercept": -0.003908087495500836,
    "n_samples": 430990
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "yv0": -4.596911198104602,
```

```
    "v": 2.4450870885452235e-05,
    "d": -0.38531195488214753,
    "vd": 2.0182017444359035,
    "v2d": -0.008059615915940201,
    "d3": -2.278843981407281,
    "vd3": 2.063936994386932,
    "v2d3": -0.24142354648405742,
    "sr": -0.013631035087834203,
    "vsr": -0.02212312590705554,
    "d_abs_d": 0.8309043641045651,
    "a_long": -4.8739748583120935e-05,
    "intercept": -0.00032466116824531 31,
    "n_samples": 615175
  },
  "HYUNDAI_IONIQ_5": {
    "yv0": 5.722836662886642,
    "v": 7.196960609318881e-05,
    "d": -0.17642093995484603,
    "vd": -1.5621262047054798,
    "v2d": -0.006411652225165284,
    "d3": -0.670272571699
```

module-2.v2_agent-07

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "K_us": 0.0033098390487113745,
    "L_eff": 4.020011778860773,
    "bias_static": -0.005395938573657566,
    "bias_steer": 0.03997187624313666,
    "bv_steer": 0.00022967560413598412
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "K_us": 0.0029726845285090037,
    "L_eff": 2.8553188283792372,
    "bias_static": 0.0007957279020199589,
    "bias_steer": 0.1222792973746867 1,
    "bv_steer": 0.02
```

```
  },
  "HYUNDAI_IONIQ_5": {
    "K_us": 0.004129560125359133,
    "L_eff": 2.8437357448117706,
    "bias_static": 0.0020383061788411035,
    "bias_steer": -0.006689089312079554,
    "bv_steer": -0.02
  },
  "TESLA_MODEL_3": {
    "_noop": 0.0
  }
}
```

module-2.v2_agent-09

```
{
  "TESLA_MODEL_3": {
    "a": 1.0,
    "K": 0.0,
    "b": 0.0,
    "shift": 0,
    "note": "Tesla sim truth column IS the V0 KS output (psi_dot_
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.93835,
    "K": 0.000885919,
    "b": -0.004447,
    "shift": 2
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 1.1781,
    "K": 0.000883249,
    "b": 0.000174,
    "shift": 2
```

```
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.93602,
    "K": 0.001005092,
    "b": 0.001992,
    "shift": 2
  }
}
```


6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

> **Clean cohort:** no agent referenced truth-derived column names inside their predict() body.

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	30	0
has_manifest_json	30	0
manifest_parsable	30	0
manifest_declares_predict_callable	30	0
manifest_declares_platform_support	30	0
has_predict_py	30	0
has_coeffs_json	24	6
has_report	6	24

8. Worst-of-cohort

Lowest yaw Δ%

- m1-agent-08 (+40.4%)
- m2-agent-04 (+44.6%)
- module-2.v2_agent-03 (+45.3%)

Lowest CTE Δ%

- m1-agent-10 (+49.5%)
- m2-agent-07 (+49.5%)
- m2-agent-04 (+50.0%)